

ENGR 102 Summer Bridge

Day 2 Student Quiz

Variables, Arithmetic, Conversion, and Formatted Output

20 points • approximately 20-25 minutes

Name		UIN		Section	
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Boolean-operator convention. Python operators appear as **and**, **or**, and **not**. Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Trace quotient, remainder, and final state.

4 points

```
minutes = 367
hours = minutes // 60
remaining = minutes % 60
code = hours * 10 + remaining
print(hours, remaining, code)
```

Show each arithmetic result and give the exact output.

Question 2. Distinguish text combination from numeric addition.

4 points

```
raw = "8"
extra = 3
print(raw + str(extra))
print(int(raw) + extra)
```

Name the type used by each + operation and give both exact output lines.

Question 3. Repair a missing input conversion.

4 points

```
kits = "4"
total = kits * 6
print(f"Total items: {total}")
```

Give the actual output, identify the stored type that caused it, and write the minimal repair.

Question 4. Program construction

8 points

Write one complete Python program that satisfies every requirement.

- Read length and width as floating-point inputs.
- Compute rectangle area and perimeter.
- Print Area: followed by one decimal place, then Perimeter: followed by one decimal place.
- The program must work for values other than the example.

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Test evidence / expected result